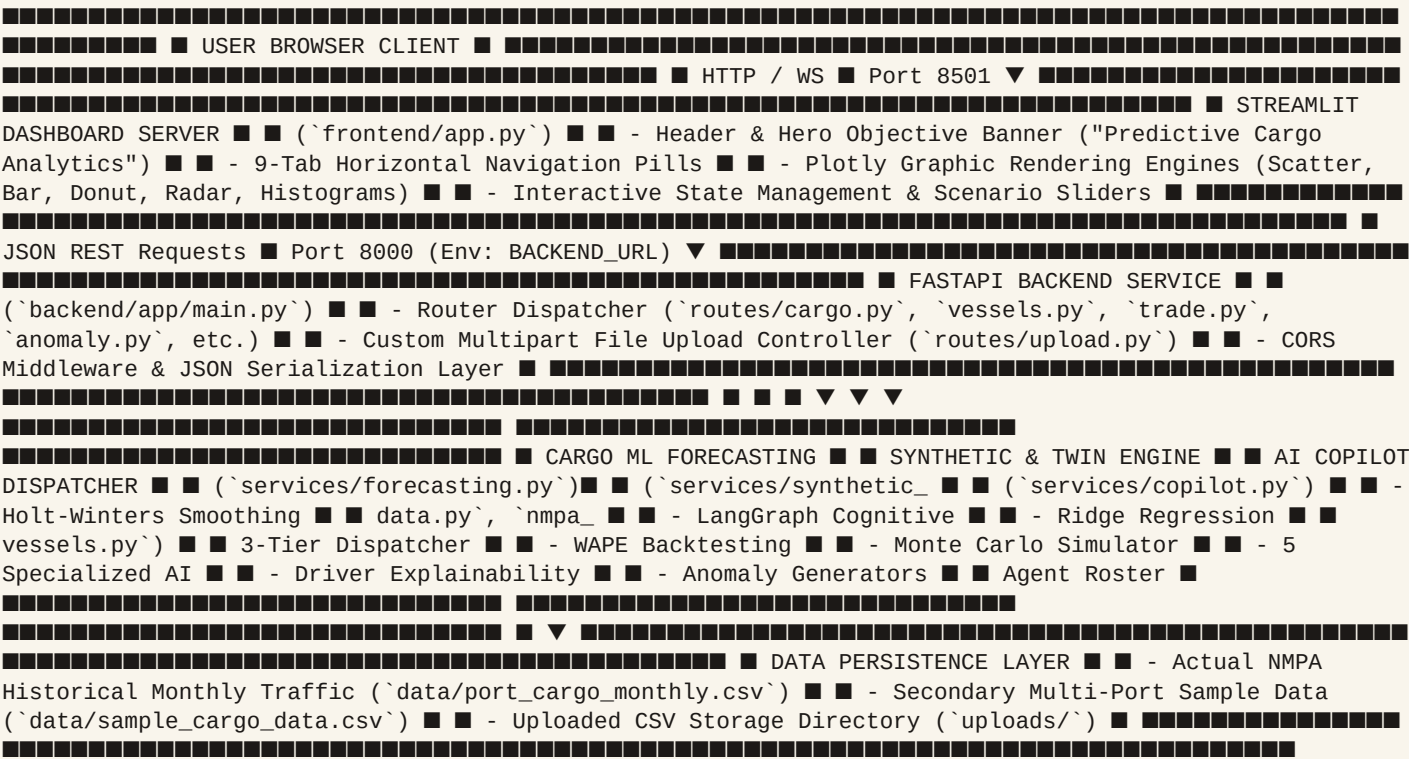


04 — System Architecture & Technical Specifications

High-Level System Architecture Diagram



1. Frontend Architecture

- **Framework:** Streamlit 1.50.0 (Python-native web application server).
- **Entry Point File:** `CODE_PH_0` (1,560+ lines of modular UI code).
- **Styling & Design Token System:**
 - Custom injected CSS stylesheet (`CODE_PH_0`).
 - Warm Amber / Cream Color Palette (`CODE_PH_0` background, `CODE_PH_1` dark neutral text, `CODE_PH_2` Gold primary accents, `CODE_PH_3` highlight cards).
- **Navigation Architecture:**
 - 9-Tab Horizontal Navigation Pills:
 1. `CODE_PH_0`
 2. `CODE_PH_0`
 3. `CODE_PH_0` (Core Priority Feature)
 4. `CODE_PH_0` (`CODE_PH_1`)
 5. `CODE_PH_0` (`CODE_PH_1`)

6. `CODE_PH_0`

7. `CODE_PH_0`

8. `CODE_PH_0`

9. `CODE_PH_0`

- **Charting Engines:** Plotly 6.7.0 (Plotly Express `CODE_PH_0` and Plotly Graph Objects `CODE_PH_1` with customized theme `CODE_PH_2`).
 - **API Communication Layer:** Custom synchronous HTTP helper functions `CODE_PH_0` and `CODE_PH_1` utilizing the `CODE_PH_2` library with configured timeout handling.
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■■ 2. Backend Architecture

- **Framework:** FastAPI 0.128.8 running on Uvicorn ASGI Server (`CODE_PH_0`).
 - **Entry Point File:** `CODE_PH_0`.
 - **Middlewares:** `CODE_PH_0` (permits cross-origin requests from Streamlit frontend).
 - **Router Modular Structure** (`CODE_PH_0`):
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`, `CODE_PH_3`, `CODE_PH_4`, `CODE_PH_5`, `CODE_PH_6`, `CODE_PH_7`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`, `CODE_PH_3`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`, `CODE_PH_3`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`, `CODE_PH_3`, `CODE_PH_4`.
 - `CODE_PH_0`: Serves `CODE_PH_1`, `CODE_PH_2`, `CODE_PH_3`, `CODE_PH_4`.
 - `CODE_PH_0`: Serves `CODE_PH_1` for custom CSV report uploads.
 - `CODE_PH_0`: Serves `CODE_PH_1` liveness checks.
-

■■ 3. Database & Data Storage Layer

- **Storage Strategy:** In-Memory Pandas DataFrame Data Pool backed by persistent CSV files.
 - **Primary Dataset:** `CODE_PH_0` (1,354 records across 2021–2026 NMPA monthly cargo operations).
 - **Secondary Datasets:** `CODE_PH_0` (multi-port benchmark data) and `CODE_PH_1` directory for dynamic user uploads.
 - **Data Schemas:**
 - `CODE_PH_0` (str), `CODE_PH_1` (int), `CODE_PH_2` (str: LOADED/UNLOADED), `CODE_PH_3` (str: Crude, Coal, Containers, etc.), `CODE_PH_4` (float), `CODE_PH_5` (float), `CODE_PH_6` (float), `CODE_PH_7` (float), `CODE_PH_8` (float).
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■ 4. AI & Machine Learning Architecture

- **Primary Time-Series Forecasting Model:**
 - **Combination Model:** Holt-Winters Exponential Smoothing + Multi-Feature Ridge Regression with L2 regularization (**CODE_PH_0**).
 - **Features:** Lagged monthly volumes (t-1, t-12), rolling 3-month stats, vessel arrival counts, seasonal indicator dummies.
 - **Prediction Bounds:** 95% dynamic confidence intervals scaled via residual standard error.
 - **Model Backtesting & Metrics Engine:**
 - **Baseline Model:** Seasonal Naive / 12-month rolling historical average.
 - **Evaluation Metrics:** Weighted Absolute Percentage Error (WAPE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE).
 - **Explainability Module:**
 - Multi-factor regression weight decomposition calculating percentage contribution of monsoon swells, hinterland demand, and trade momentum.
 - **Simulation & Anomaly Algorithms:**
 - Isolation Forest, LSTM, Autoencoder, and Transformer simulation models for anomaly detection (**CODE_PH_0**).
 - Monte Carlo 1,000-iteration probability distribution generator for trade incentive revenue modeling.
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■ 5. External Services & APIs

- **Current Implementation:** The application is fully self-contained with simulated live feeds for external data providers (GDELT trade intelligence feeds, Bloomberg commodity price indices, MITRE ATT&CK security threat framework, and NMPA AIS vessel coordinate feeds).